

CPS 200

Lecture 03

Jan 19, 2026

Kyu Hur

Schedule

Date	Session	Quiz	Topic
Jan 5			Reading Day
Jan 7	1	No	Intro to Programming I
Jan 12			National Holiday
Jan 14	2	No	Think Python: Programming, Variables and Statements
Jan 19	3	Yes	Think Python: Variables and Statements, Functions
Jan 21	4	Yes	Think Python: Functions and Interfaces, Conditionals and Recursion
Jan 26	5	Yes	Think Python: Conditionals and Recursion, Return Values
Jan 28	6	Yes	Think Python: Iteration and Search, Strings and Regular Expressions
Feb 2	7	Yes	Think Python: Strings and Regular Expressions, Lists
Feb 4	8	Yes	Think Python: Dictionaries, Tuples
Feb 9	9	No	Midterm
Feb 11			National Holiday
Feb 16	10	Yes	Think Python: Tuples, Text Analysis and Generation
Feb 18	11	Yes	Think Python: Files and Databases, Classes and Functions
Feb 23			National Holiday
Feb 25	12	Yes	Think Python: Classes and Functions, Classes and Methods
Mar 2	13	Yes	Think Python: Classes and Objects, Inheritance
Mar 4	14	Yes	Open Data Structures: Introduction & Array-Based Lists
Mar 9	15	Yes	Open Data Structures: Linked Lists
Mar 11	16	Yes	Open Data Structures: Hash Tables
Mar 16	17	Yes	Open Data Structures: Binary Trees
Mar 18	18	Yes	Open Data Structures: Sorting Algorithms & Group Project
Mar 23	19	Yes	Open Data Structures: Graphs & Group Project
Mar 25	20	No	Group Project
Mar 30	21	No	Group Project
Apr 1	22	No	Final Exam (Tentative)

Schedule

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Jan 7	1	No	Intro to Programming I
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Jan 14	2	No	Think Python: Programming as a way of thinking, Variables and Statements
Jan 19	3	Yes	Think Python: Variables and Statements, Functions
Jan 21	4	Yes	Think Python: Functions and Interfaces, Conditionals and Recursion
Jan 26	5	Yes	Think Python: Conditionals and Recursion, Return Values
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Feb 4	8	Yes	Think Python: Dictionaries, Tuples

From Lecture 02

Review

- What is a **type** in Python?
- Can you name some data types in Python? Please speak to your partners and provide them with examples.

Review

- Which of the following is an **integer** type?
 - a. 3.0
 - b. 4
 - c. "4"
 - d. "b"

Review

- Which of the following is a **string** type?
 - a. b
 - b. "4"
 - c. 3.0
 - d. 4

Review

- Which of the following is a **float** type?
 - a. 3.0
 - b. 3
 - c. "3"
 - d. "3.0"

Review

- What does the function **type()** do? For example, what should be expect when we run the code **type(3)**? What is the output?
- How about **type("b")**? **type(1.0)**?

Review

- What does the function **len()** do? What would be the output of **len("hello")**? How about **len(3.0)**?

Review

- What happens if you run the function **int()** on a number? For example, **int(3.0)** produces what output?
- How about **int("b")**? What happens here?
- Explain what **str()** and **float()** does to your partners.

Lecture 03

The **print()** function

- The **print()** function is explicitly used to print a value
- Jupyter notebooks (secretly) print the last value in a code cell
- In your code, you should be more explicitly use **print()** to output the value you want to see, especially as you write more complex code
 - a. `print("Hi")`
 - b. `print(6)`
 - c. `print(x)`

Importing

- Python comes with a base set of **variables** and **functions** that do certain basic operations
 - a. Can you name a few that we learned last time? What are they used for?
 - b. **len()**, **int()**, **str()**, **float()**, **type()**

Importing

- Other variables and functions, we need to import them
- We call these **libraries** or **modules**
 - a. **math** is a library / module that allows access to more variables and functions
 - b. **import** is the keyword in Python (e.g. `import math`)

Importing

- Other libraries and modules developed by the community
 - a. Libraries that can be imported right away e.g. **math**
 - b. Libraries that need to be downloaded THEN imported e.g. **tensorflow** (google), **pytorch** (facebook), etc
 - c. Out of scope for this class

Importing

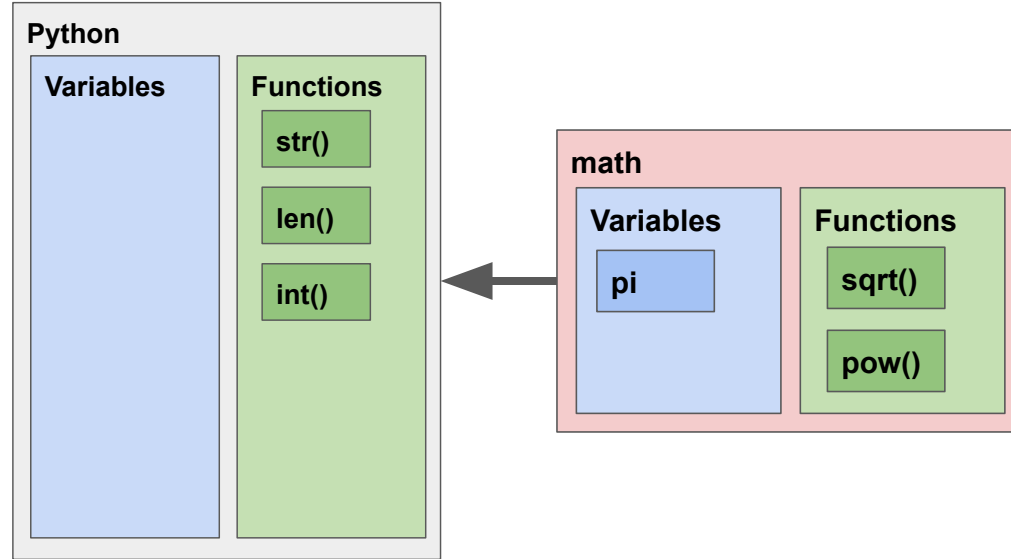
- With **math**, we have access to the **pi** variable
 - a. We access it by using a period in between the library and the variable, e.g.
 - **math.pi**
 - If we do **print(math.pi)**, what will we get?

Importing

- We also have access to the functions under math, like **sqrt** or **pow**
 - a. We access these functions like so:
 - **math.sqrt()**
 - **math.pow()**
 - If we do **print(math.sqrt(16))** what will we get?

Importing

- **import** loads stuff from storage (SSD/HDD) into memory (RAM)
- It does NOT download the **math** library from the internet (it is already downloaded)
- On the right is a diagram of running `import math`



Think Python

Think Python: Variables and Statements

- [Chapter 02.ipynb](#)

Think Python: Functions

- [Chapter 03.ipynb](#)

End